

GED Mathematics Solutions Slope and Equations of Lines

Complete Answer Key with Step-by-Step Explanations

Contents

1 Part A: Calculating Slope from Two Points (Questions 1-9)

Question 1: Find the slope between $(2, 3)$ and $(5, 9)$

Solution:

Using the slope formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$m = \frac{9 - 3}{5 - 2}$$

$$m = \frac{6}{3}$$

$$m = 2$$

Answer: $m = 2$

The slope is positive, indicating the line rises from left to right.

Question 2: Find the slope between $(-1, 4)$ and $(3, 2)$

Solution:

$$m = \frac{2 - 4}{3 - (-1)}$$

$$m = \frac{-2}{4}$$

$$m = -\frac{1}{2}$$

Answer: $m = -\frac{1}{2}$

The negative slope indicates the line falls from left to right.

Question 3: Find the slope between $(0, 0)$ and $(4, 8)$

Solution:

$$m = \frac{8 - 0}{4 - 0}$$

$$m = \frac{8}{4}$$

$$m = 2$$

Answer: $m = 2$

This line passes through the origin with a slope of 2.

Question 4: Find the slope between $(-3, -2)$ and $(1, 6)$

Solution:

$$m = \frac{6 - (-2)}{1 - (-3)}$$

$$m = \frac{6 + 2}{1 + 3}$$

$$m = \frac{8}{4}$$

$$m = 2$$

Answer: $m = 2$

Question 5: Find the slope between $(5, 1)$ and $(5, 7)$

Solution:

$$m = \frac{7 - 1}{5 - 5}$$

$$m = \frac{6}{0}$$

Answer: The slope is **undefined**.

Both points have the same x -coordinate, so the line is vertical. Vertical lines have undefined slope.

Question 6: Find the slope between $(2, -3)$ and $(8, -3)$

Solution:

$$m = \frac{-3 - (-3)}{8 - 2}$$

$$m = \frac{0}{6}$$

$$m = 0$$

Answer: $m = 0$

Both points have the same y -coordinate, so the line is horizontal. Horizontal lines have zero slope.

Question 7: Find the slope between $(-4, 3)$ and $(2, -9)$

Solution:

$$m = \frac{-9 - 3}{2 - (-4)}$$

$$m = \frac{-12}{6}$$

$$m = -2$$

Answer: $m = -2$

Question 8: Find the slope between $(0, 5)$ and $(-3, 2)$

Solution:

$$m = \frac{2 - 5}{-3 - 0}$$

$$m = \frac{-3}{-3}$$

$$m = 1$$

Answer: $m = 1$

Question 9: Find the slope between $(-2, -5)$ and $(4, 7)$

Solution:

$$m = \frac{7 - (-5)}{4 - (-2)}$$

$$m = \frac{7 + 5}{4 + 2}$$

$$m = \frac{12}{6}$$

$$m = 2$$

Answer: $m = 2$

2 Part B: Identifying Parallel and Perpendicular Lines (Questions 10-17)

Key Concepts:

- **Parallel lines** have equal slopes: $m_1 = m_2$
- **Perpendicular lines** have slopes that are negative reciprocals: $m_1 \cdot m_2 = -1$ or $m_2 = -\frac{1}{m_1}$

Question 10: $m_1 = 3$ and $m_2 = 3$

Solution:

Since $m_1 = m_2 = 3$, the lines are **parallel**.

Question 11: $m_1 = 2$ and $m_2 = -\frac{1}{2}$

Solution:

Check if perpendicular: $m_1 \cdot m_2 = 2 \cdot \left(-\frac{1}{2}\right) = -1$

The lines are **perpendicular**.

Question 12: $m_1 = \frac{3}{4}$ and $m_2 = \frac{3}{4}$

Solution:

Since $m_1 = m_2 = \frac{3}{4}$, the lines are **parallel**.

Question 13: $m_1 = 5$ and $m_2 = -5$

Solution:

Check if perpendicular: $m_1 \cdot m_2 = 5 \cdot (-5) = -25 \neq -1$

The slopes are not equal and not negative reciprocals, so the lines are **neither parallel nor perpendicular**.

Question 14: $m_1 = -\frac{2}{3}$ and $m_2 = \frac{3}{2}$

Solution:

Check if perpendicular: $m_1 \cdot m_2 = \left(-\frac{2}{3}\right) \cdot \frac{3}{2} = -\frac{6}{6} = -1$

The lines are **perpendicular**.

Question 15: $m_1 = \frac{1}{4}$ and $m_2 = -4$

Solution:

Check if perpendicular: $m_1 \cdot m_2 = \frac{1}{4} \cdot (-4) = -1$

The lines are **perpendicular**.

Question 16: $m_1 = -1$ and $m_2 = 1$

Solution:

Check if perpendicular: $m_1 \cdot m_2 = (-1) \cdot 1 = -1$

The lines are **perpendicular**.

Question 17: $m_1 = \frac{5}{6}$ and $m_2 = \frac{6}{5}$

Solution:

Check if perpendicular: $m_1 \cdot m_2 = \frac{5}{6} \cdot \frac{6}{5} = 1 \neq -1$

The slopes are not equal and not negative reciprocals, so the lines are **neither parallel nor perpendicular**.

3 Part C: Finding Equations of Lines (Questions 18-27)

Key Form: Point-Slope Form

$$y - y_1 = m(x - x_1)$$

This can be converted to slope-intercept form: $y = mx + b$

Question 18: Find the equation of a line with slope $m = 2$ passing through $(1, 3)$

Solution:

Using point-slope form:

$$y - 3 = 2(x - 1)$$

$$y - 3 = 2x - 2$$

$$y = 2x + 1$$

Answer: $y = 2x + 1$

Question 19: Find the equation of a line with slope $m = -3$ passing through $(2, -5)$

Solution:

$$y - (-5) = -3(x - 2)$$

$$y + 5 = -3x + 6$$

$$y = -3x + 1$$

Answer: $y = -3x + 1$

Question 20: Find the equation of a line passing through $(1, 2)$ and $(3, 6)$

Solution:

First, find the slope:

$$m = \frac{6 - 2}{3 - 1} = \frac{4}{2} = 2$$

Using point-slope form with $(1, 2)$:

$$y - 2 = 2(x - 1)$$

$$y - 2 = 2x - 2$$

$$y = 2x$$

Answer: $y = 2x$

Question 21: Find the equation of a line with slope $m = \frac{1}{2}$ passing through $(0, 4)$

Solution:

Using point-slope form:

$$y - 4 = \frac{1}{2}(x - 0)$$

$$y - 4 = \frac{1}{2}x$$

$$y = \frac{1}{2}x + 4$$

Answer: $y = \frac{1}{2}x + 4$

Note: The y -intercept is 4 since the point $(0, 4)$ lies on the y -axis.

Question 22: Find the equation of a line passing through $(-2, 3)$ and $(4, 6)$

Solution:

First, find the slope:

$$m = \frac{6 - 3}{4 - (-2)} = \frac{3}{6} = \frac{1}{2}$$

Using point-slope form with $(4, 6)$:

$$y - 6 = \frac{1}{2}(x - 4)$$

$$y - 6 = \frac{1}{2}x - 2$$

$$y = \frac{1}{2}x + 4$$

Answer: $y = \frac{1}{2}x + 4$

Question 23: Find the equation of a line with slope $m = -\frac{2}{3}$ passing through $(3, 1)$

Solution:

$$\begin{aligned}y - 1 &= -\frac{2}{3}(x - 3) \\y - 1 &= -\frac{2}{3}x + 2 \\y &= -\frac{2}{3}x + 3\end{aligned}$$

Answer: $y = -\frac{2}{3}x + 3$

Question 24: Find the equation of a line passing through $(0, 0)$ and $(5, 10)$

Solution:

First, find the slope:

$$m = \frac{10 - 0}{5 - 0} = \frac{10}{5} = 2$$

Using point-slope form with $(0, 0)$:

$$\begin{aligned}y - 0 &= 2(x - 0) \\y &= 2x\end{aligned}$$

Answer: $y = 2x$

This is a line through the origin.

Question 25: Find the equation of a line with slope $m = 0$ passing through $(2, 5)$

Solution:

$$\begin{aligned}y - 5 &= 0(x - 2) \\y - 5 &= 0 \\y &= 5\end{aligned}$$

Answer: $y = 5$

This is a horizontal line at $y = 5$.

Question 26: Find the equation of a line passing through $(-3, -1)$ and $(2, 4)$

Solution:

First, find the slope:

$$m = \frac{4 - (-1)}{2 - (-3)} = \frac{5}{5} = 1$$

Using point-slope form with $(2, 4)$:

$$y - 4 = 1(x - 2)$$

$$y - 4 = x - 2$$

$$y = x + 2$$

Answer: $y = x + 2$

Question 27: Find the equation of a line with slope $m = -1$ passing through $(4, -2)$

Solution:

$$y - (-2) = -1(x - 4)$$

$$y + 2 = -x + 4$$

$$y = -x + 2$$

Answer: $y = -x + 2$

4 Part D: Graphing Linear Equations (Questions 28-33)

Key Concept: Slope-Intercept Form

$$y = mx + b$$

where m is the slope and b is the y -intercept.

Question 28: $2x + y = 5$

Solution:

Solve for y :

$$\begin{aligned} 2x + y &= 5 \\ y &= -2x + 5 \end{aligned}$$

Slope: $m = -2$

y -intercept: $b = 5$ (point: $(0, 5)$)

Question 29: $3x - y = 6$

Solution:

Solve for y :

$$\begin{aligned} 3x - y &= 6 \\ -y &= -3x + 6 \\ y &= 3x - 6 \end{aligned}$$

Slope: $m = 3$

y -intercept: $b = -6$ (point: $(0, -6)$)

Question 30: $4x + 2y = 8$

Solution:

Solve for y :

$$\begin{aligned} 4x + 2y &= 8 \\ 2y &= -4x + 8 \\ y &= -2x + 4 \end{aligned}$$

Slope: $m = -2$

y -intercept: $b = 4$ (point: $(0, 4)$)

Question 31: $y = 3x - 2$

Solution:

This equation is already in slope-intercept form.

Slope: $m = 3$

y-intercept: $b = -2$ (point: $(0, -2)$)

Question 32: $-2x + 3y = 9$

Solution:

Solve for y :

$$-2x + 3y = 9$$

$$3y = 2x + 9$$

$$y = \frac{2}{3}x + 3$$

Slope: $m = \frac{2}{3}$

y-intercept: $b = 3$ (point: $(0, 3)$)

Question 33: $5x - 5y = 10$

Solution:

Solve for y :

$$5x - 5y = 10$$

$$-5y = -5x + 10$$

$$y = x - 2$$

Slope: $m = 1$

y-intercept: $b = -2$ (point: $(0, -2)$)

5 Part E: Real-World Applications of Slope (Questions 34-38)

Question 34: A wheelchair ramp rises 2 feet over a horizontal distance of 24 feet. What is the slope of the ramp?

Solution:

Slope is defined as the ratio of rise (vertical change) to run (horizontal change).

$$\begin{aligned}\text{Slope} &= \frac{\text{rise}}{\text{run}} \\ &= \frac{2}{24} \\ &= \frac{1}{12}\end{aligned}$$

Answer: $m = \frac{1}{12} \approx 0.083$ or about 8.3%

The slope is very gentle, which is appropriate for wheelchair accessibility (ADA standard is typically 1:12).

Question 35: A staircase has steps that rise 7 inches for every 10 inches of horizontal distance. What is the slope of the staircase?

Solution:

$$\begin{aligned}\text{Slope} &= \frac{\text{rise}}{\text{run}} \\ &= \frac{7}{10} \\ &= 0.7\end{aligned}$$

Answer: $m = \frac{7}{10}$ or $m = 0.7$

Question 36: A road elevation increases 300 feet for every 2000 feet of horizontal distance. Express this as a slope in simplest form.

Solution:

$$\text{Slope} = \frac{300}{2000}$$

Simplify by finding the GCD of 300 and 2000:

$$\frac{300}{2000} = \frac{3}{20}$$

Answer: $m = \frac{3}{20}$ or $m = 0.15$ or 15%

Question 37: A mountain trail descends 400 feet in elevation over a horizontal distance of 1 mile (5280 feet). What is the slope of the trail?

Solution:

Since the trail descends (goes down), the slope is negative.

$$\text{Slope} = -\frac{400}{5280}$$

Simplify by finding the GCD of 400 and 5280:

$$-\frac{400}{5280} = -\frac{5}{66}$$

Answer: $m = -\frac{5}{66} \approx -0.0758$ or about -7.58%

Question 38: A ladder leans against a wall, reaching 12 feet high. The base of the ladder is 5 feet from the wall. What is the slope of the ladder?

Solution:

The ladder reaches from the point $(0, 0)$ (base on ground) to the point $(5, 12)$ (top of wall).

$$\begin{aligned}\text{Slope} &= \frac{12 - 0}{5 - 0} \\ &= \frac{12}{5} \\ &= 2.4\end{aligned}$$

Answer: $m = \frac{12}{5}$ or $m = 2.4$

The slope is steep because the ladder rises much more than it extends horizontally.

6 Part F: Distance Between Points (Questions 39-43)

Distance Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Question 39: Find the distance between $(0, 0)$ and $(3, 4)$

Solution:

$$\begin{aligned} d &= \sqrt{(3 - 0)^2 + (4 - 0)^2} \\ &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

Answer: $d = 5$ units

This is the famous 3-4-5 Pythagorean triple.

Question 40: Find the distance between $(1, 2)$ and $(4, 6)$

Solution:

$$\begin{aligned} d &= \sqrt{(4 - 1)^2 + (6 - 2)^2} \\ &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

Answer: $d = 5$ units

Question 41: Find the distance between $(-2, 1)$ and $(3, 5)$

Solution:

$$\begin{aligned} d &= \sqrt{(3 - (-2))^2 + (5 - 1)^2} \\ &= \sqrt{(5)^2 + (4)^2} \\ &= \sqrt{25 + 16} \\ &= \sqrt{41} \end{aligned}$$

Answer: $d = \sqrt{41} \approx 6.4$ units

Question 42: Find the distance between $(0, -3)$ and $(4, 0)$

Solution:

$$\begin{aligned}d &= \sqrt{(4 - 0)^2 + (0 - (-3))^2} \\&= \sqrt{4^2 + 3^2} \\&= \sqrt{16 + 9} \\&= \sqrt{25} \\&= 5\end{aligned}$$

Answer: $d = 5$ units

Another 3-4-5 Pythagorean triple $(3, 4, 5)$.

Question 43: Find the distance between $(-1, -1)$ and $(2, 3)$

Solution:

$$\begin{aligned}d &= \sqrt{(2 - (-1))^2 + (3 - (-1))^2} \\&= \sqrt{(3)^2 + (4)^2} \\&= \sqrt{9 + 16} \\&= \sqrt{25} \\&= 5\end{aligned}$$

Answer: $d = 5$ units

Yet another 3-4-5 Pythagorean triple.

7 Part G: Systems of Equations (Questions 44-48)

Methods: Substitution or Elimination

Question 44:
$$\begin{cases} y = 2x + 1 \\ y = -x + 4 \end{cases}$$

Solution (Substitution Method):

Since both expressions equal y , set them equal to each other:

$$2x + 1 = -x + 4$$

$$3x + 1 = 4$$

$$3x = 3$$

$$x = 1$$

Substitute $x = 1$ into the first equation:

$$y = 2(1) + 1$$

$$y = 3$$

Answer: $(x, y) = (1, 3)$

Question 45:
$$\begin{cases} x + y = 5 \\ x - y = 1 \end{cases}$$

Solution (Elimination Method):

Add the two equations to eliminate y :

$$(x + y) + (x - y) = 5 + 1$$

$$2x = 6$$

$$x = 3$$

Substitute $x = 3$ into the first equation:

$$3 + y = 5$$

$$y = 2$$

Answer: $(x, y) = (3, 2)$

Question 46: $\begin{cases} 2x + y = 7 \\ x - y = 2 \end{cases}$

Solution (Elimination Method):

Add the two equations to eliminate y :

$$(2x + y) + (x - y) = 7 + 2$$

$$3x = 9$$

$$x = 3$$

Substitute $x = 3$ into the second equation:

$$3 - y = 2$$

$$-y = -1$$

$$y = 1$$

Answer: $(x, y) = (3, 1)$

Question 47: $\begin{cases} 3x - 2y = 4 \\ x + y = 5 \end{cases}$

Solution (Substitution Method):

From the second equation, solve for y :

$$x + y = 5$$

$$y = 5 - x$$

Substitute into the first equation:

$$3x - 2(5 - x) = 4$$

$$3x - 10 + 2x = 4$$

$$5x - 10 = 4$$

$$5x = 14$$

$$x = \frac{14}{5}$$

Substitute back to find y :

$$y = 5 - \frac{14}{5}$$

$$y = \frac{25}{5} - \frac{14}{5}$$

$$y = \frac{11}{5}$$

Answer: $(x, y) = \left(\frac{14}{5}, \frac{11}{5}\right)$ or $(2.8, 2.2)$

Question 48: $\begin{cases} y = \frac{1}{2}x + 3 \\ y = -2x + 8 \end{cases}$

Solution (Substitution Method):

Set the two expressions for y equal:

$$\frac{1}{2}x + 3 = -2x + 8$$

$$\frac{1}{2}x + 2x = 8 - 3$$

$$\frac{1}{2}x + \frac{4}{2}x = 5$$

$$\frac{5}{2}x = 5$$

$$x = 2$$

Substitute $x = 2$ into the first equation:

$$y = \frac{1}{2}(2) + 3$$

$$y = 1 + 3$$

$$y = 4$$

Answer: $(x, y) = (2, 4)$

8 Part H: Mixed Review (Questions 49-50)

Question 49: Two parallel lines have equations $y = 3x + 2$ and $y = 3x - 5$. Find the perpendicular slope to these lines, and write an equation of a line that passes through $(2, 4)$ with that perpendicular slope.

Solution:

The parallel lines have slope $m = 3$.

For a perpendicular line, the slope is the negative reciprocal:

$$m_{\perp} = -\frac{1}{3}$$

Now write the equation of a line with slope $m = -\frac{1}{3}$ passing through $(2, 4)$:

$$\begin{aligned}y - 4 &= -\frac{1}{3}(x - 2) \\y - 4 &= -\frac{1}{3}x + \frac{2}{3} \\y &= -\frac{1}{3}x + \frac{2}{3} + 4 \\y &= -\frac{1}{3}x + \frac{2}{3} + \frac{12}{3} \\y &= -\frac{1}{3}x + \frac{14}{3}\end{aligned}$$

Answer:

- Perpendicular slope: $m = -\frac{1}{3}$
- Equation: $y = -\frac{1}{3}x + \frac{14}{3}$

Question 50: A line passes through the points $(-2, 1)$ and $(4, 7)$. Find the slope, write the equation in slope-intercept form, and find the distance between the two points.

Solution:

Part 1: Find the slope

$$\begin{aligned}m &= \frac{7 - 1}{4 - (-2)} \\&= \frac{6}{6} \\&= 1\end{aligned}$$

Part 2: Write the equation in slope-intercept form

Using point-slope form with point $(4, 7)$:

$$y - 7 = 1(x - 4)$$

$$y - 7 = x - 4$$

$$y = x + 3$$

Part 3: Find the distance between the two points

$$\begin{aligned} d &= \sqrt{(4 - (-2))^2 + (7 - 1)^2} \\ &= \sqrt{(6)^2 + (6)^2} \\ &= \sqrt{36 + 36} \\ &= \sqrt{72} \\ &= \sqrt{36 \cdot 2} \\ &= 6\sqrt{2} \end{aligned}$$

Answer:

- Slope: $m = 1$
- Equation: $y = x + 3$
- Distance: $d = 6\sqrt{2} \approx 8.49$ units